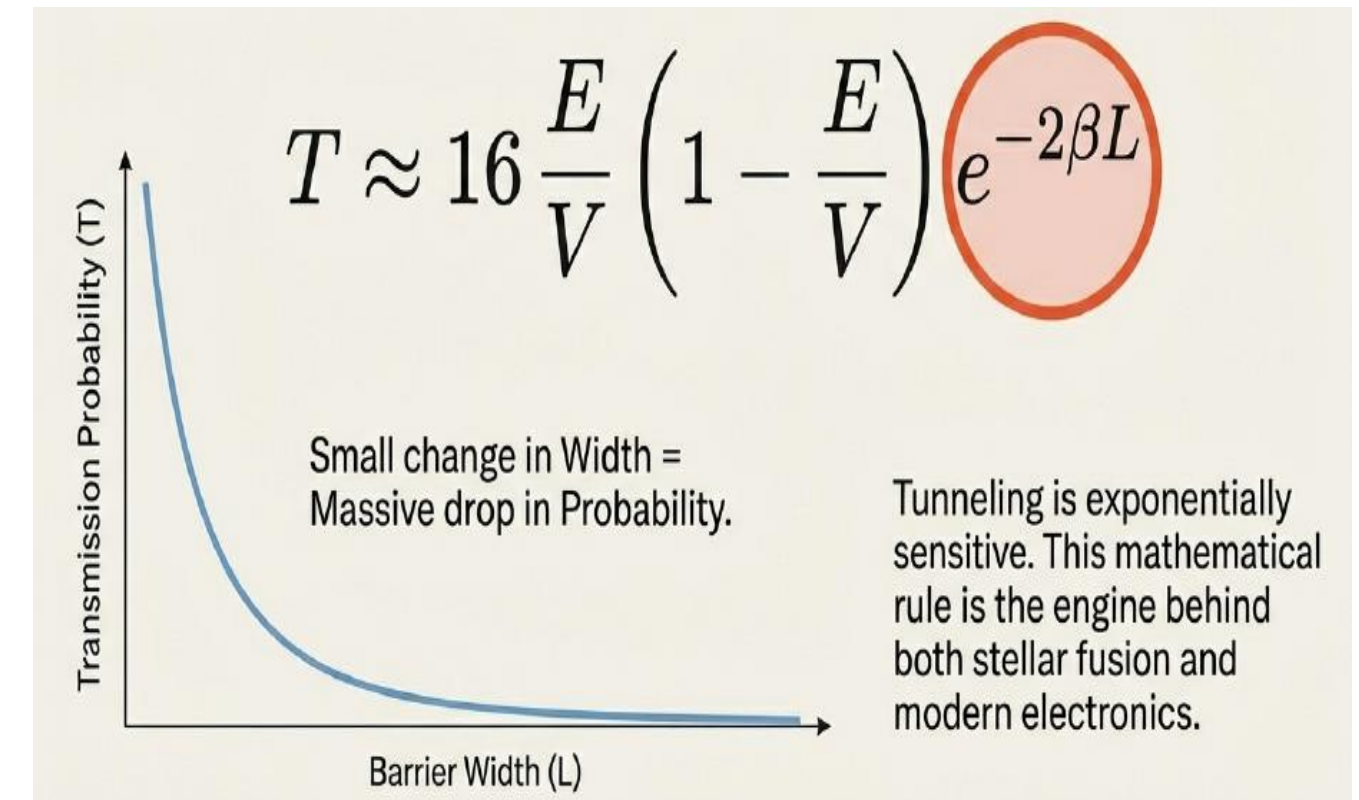
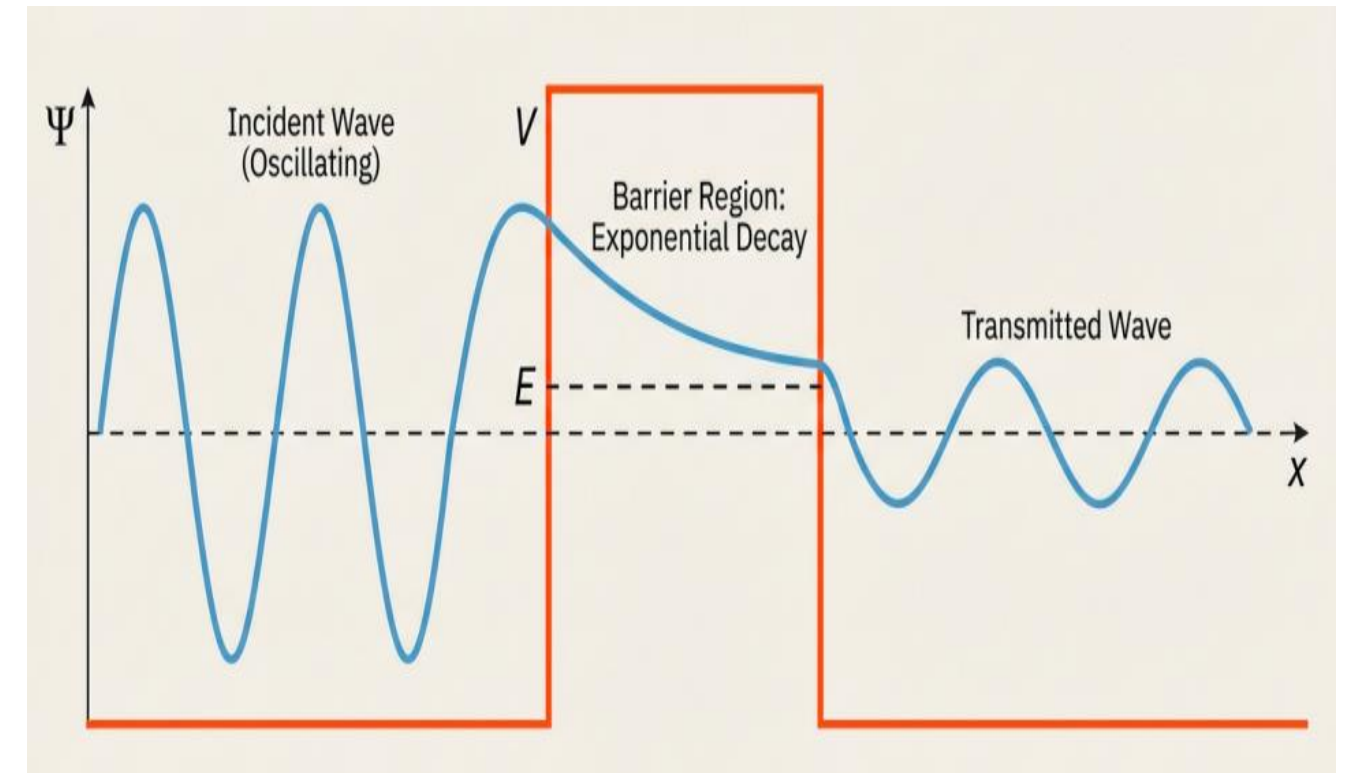


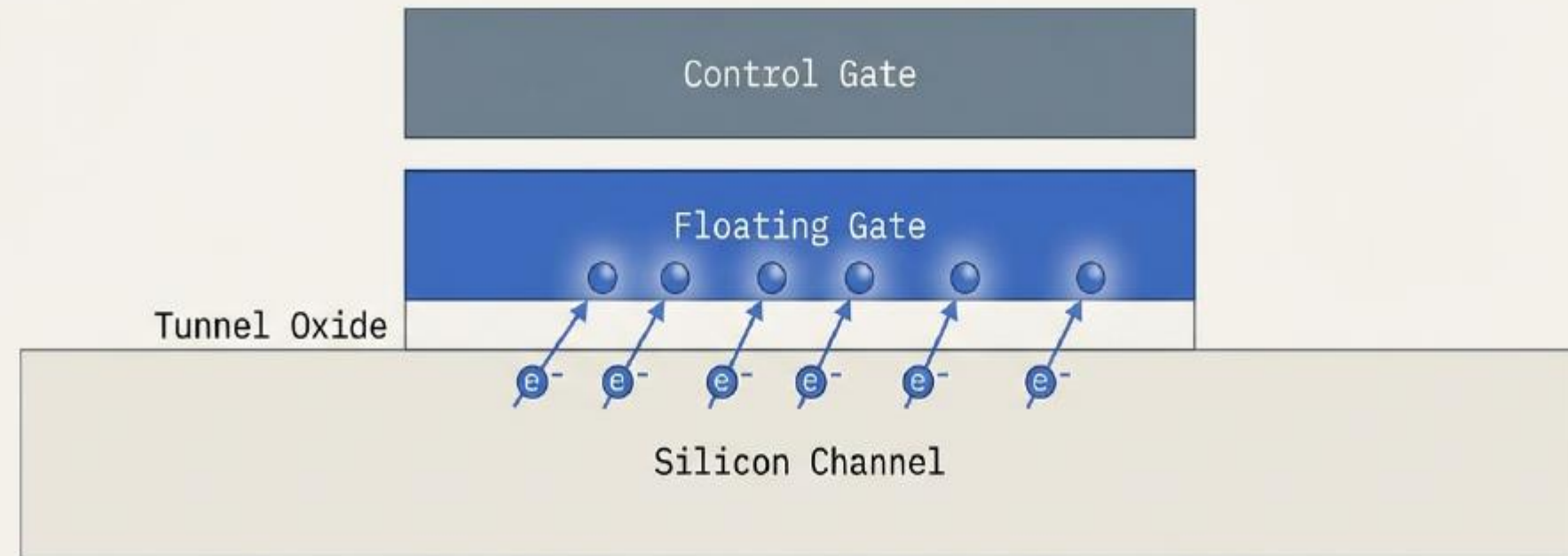
Quantum Tunneling

- In the quantum mechanics, When an electron hits a "wall" (an energy barrier) that it technically doesn't have enough energy to cross, there is still a small probability that it will simply appear on the other side.
- It doesn't "break" the barrier; it essentially leaks through it. This is called **"Tunneling"**.
- The wave decays exponentially inside the barrier.
- Inside the barrier: The particle's energy (E) is less than the potential energy (V) of the barrier. In quantum physics, the wave function persists but drops off very quickly.
- If the barrier is thin enough, the wave function doesn't reach zero before it hits the other side. A tiny portion of the wave **"pass through"** to the other side, meaning there is a non-zero chance the particle will be detected there.
- **The "Sensitivity" of Tunneling:**
 - As shown in the formula ,**the probability T** is an exponential function of the barrier width (L).
 - If you double the width of a barrier, the chance of tunneling doesn't just cut in half—it might drop by a factor of 1,000 or 1,000,000.



Tunneling application

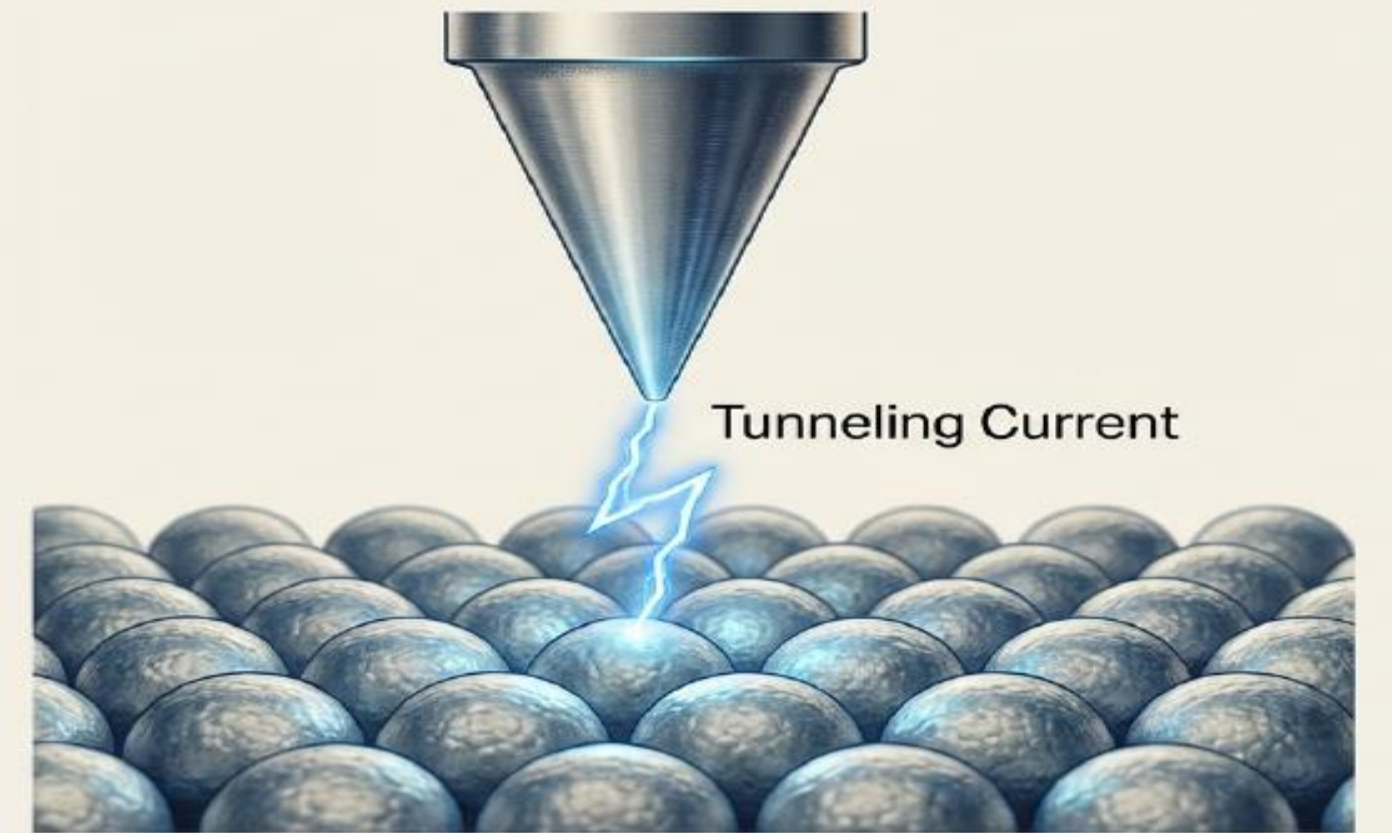
Flash Memory: The Electron Trap



Flash Memory (USB Drives & SSDs)

- Your files are stored using tunneling.
- In a Flash drive, electrons are pushed through an insulating layer onto a "floating gate" via Fowler-Nordheim tunneling.
- Once they tunnel through, they are trapped there, representing a "1" or a "0." Because the barrier is thick enough to prevent accidental tunneling back out, your data stays saved even when the power is off.

STM: Seeing the Invisible



Scanning Tunneling Microscope (STM)

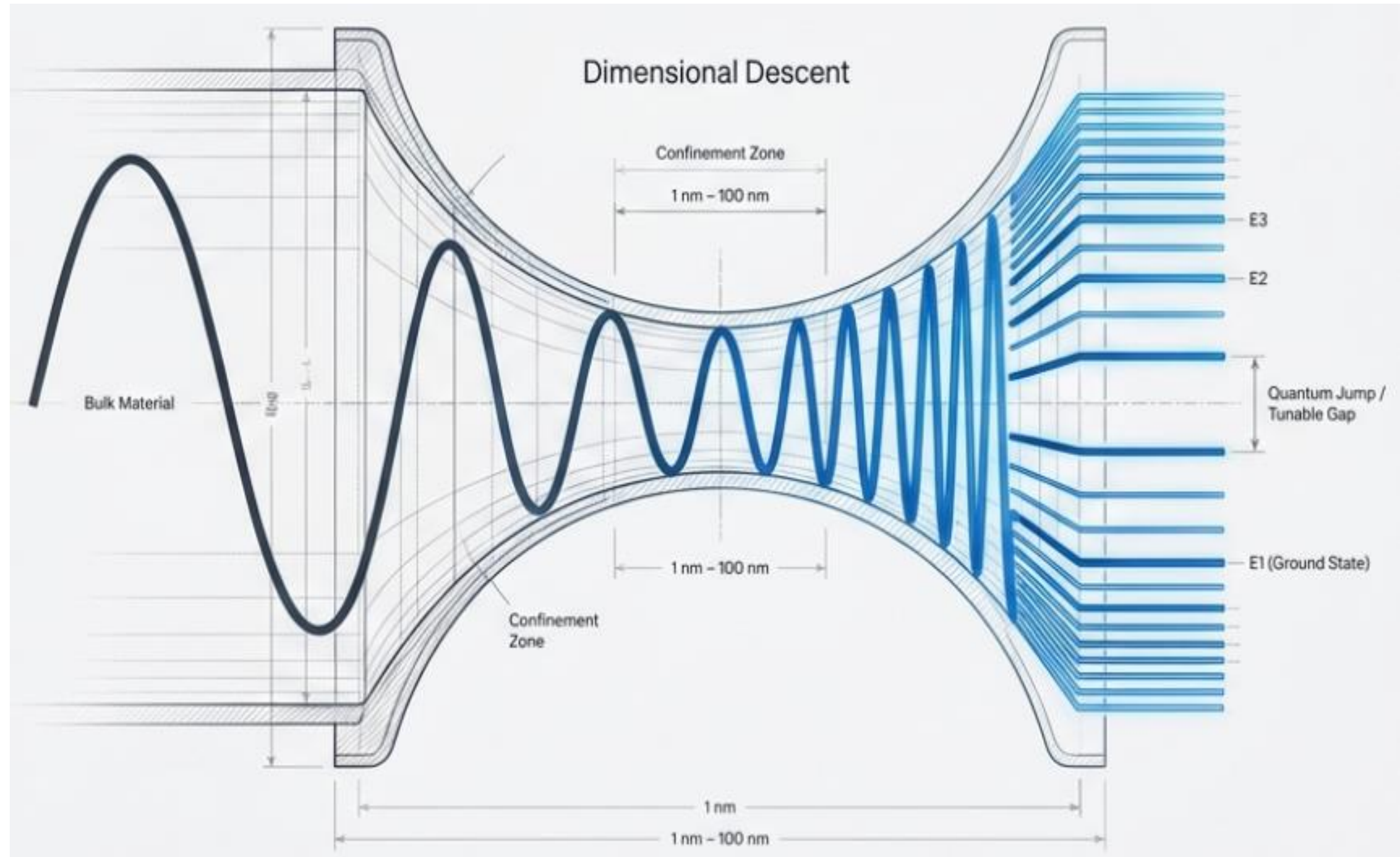
A super-sharp needle hovers just above a surface (without touching it). A small voltage is applied, and electrons tunnel across the gap.

Because tunneling is so sensitive to distance, even a change the size of a single atom changes the current flow significantly. The microscope maps these current changes to create a 3D image of the atomic surface i.e **We can see "ATOMs"**

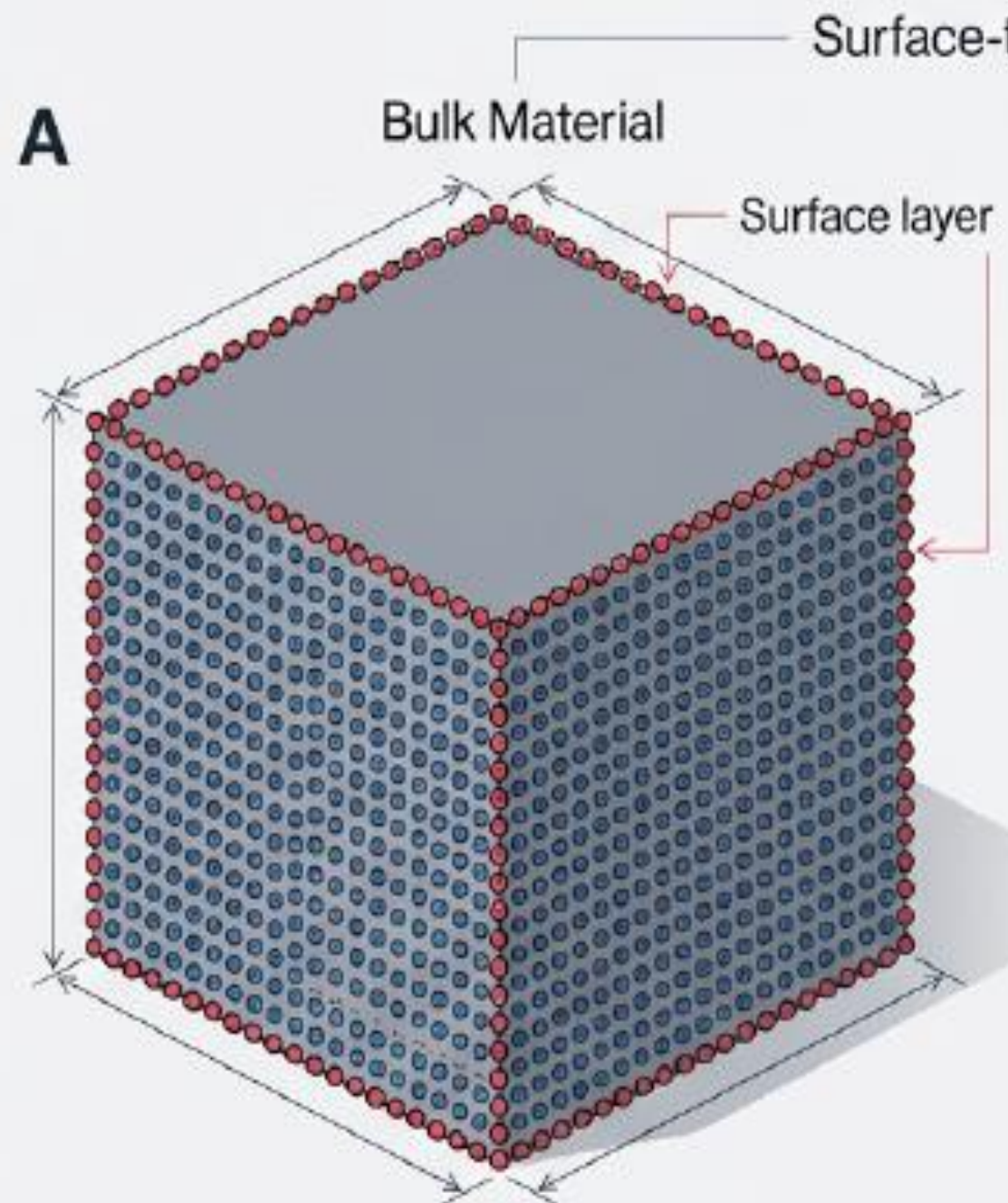
The Quantum Mechanics of Nanoparticles

From Dimensional Confinement to Scalable Computing Architectures

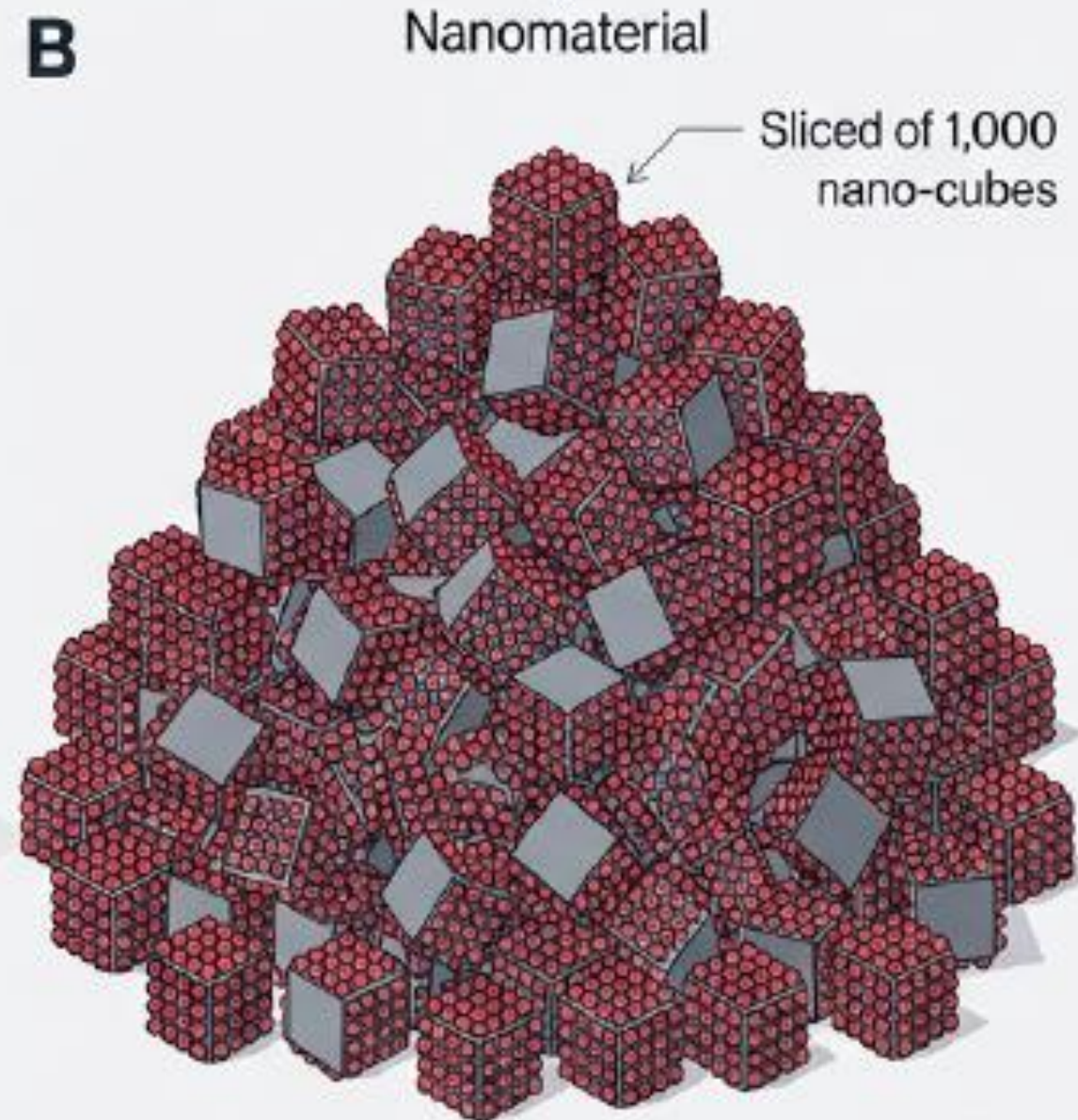
- In the macroscopic world, properties are fixed. Gold is yellow; silicon is an insulator. But when we cross the threshold into the nanoscale (1–100 nm), the rules of classical physics collapse.
- This is a story of Confinement leading to Freedom. By squeezing matter into fewer dimensions, we liberate tunable properties—color, conductivity, and logic—that define the future of technology.



The Nanoscale Threshold: Where Surface Dominates Core



Internal atoms dominate properties.



Surface atoms dominate reactivity.

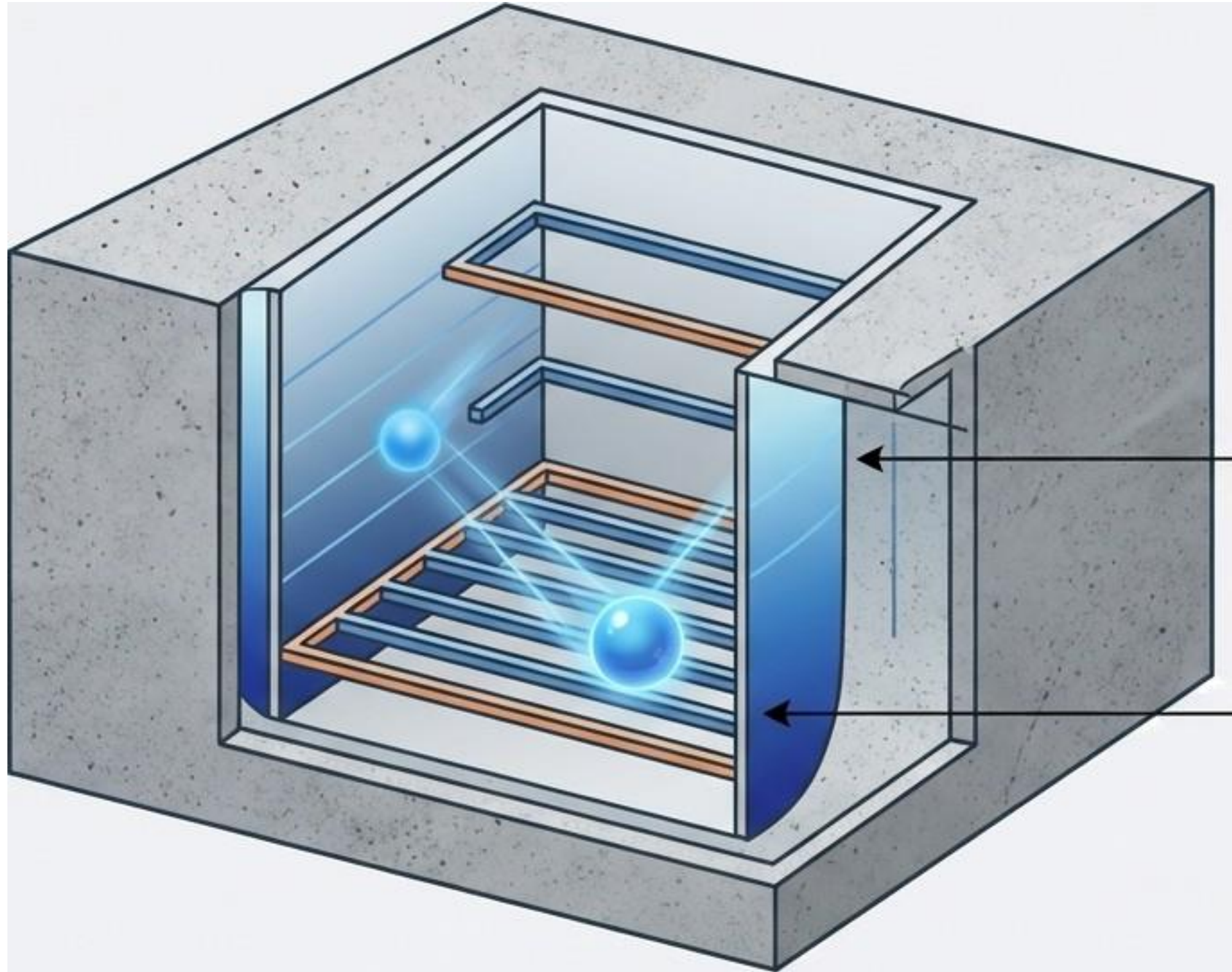
Key Concept: The Young-Laplace Law

As the particle radius (R) decreases, internal pressure (ΔP) increases inversely.

$$\Delta P \propto \frac{2\gamma}{R}$$

Result: Immense surface pressure modifies interatomic interactions and alters the bandgap.

The Physics of Confinement: The Particle in a Box



Quantum Potential Well

Core Mechanism

- **Mechanism:** An electron is trapped inside a very small region with rigid boundaries. Like , in Quantum Mechanics, When a material's size becomes smaller than the **exciton Bohr radius** (the typical electron-hole pair separation), its energy levels become quantized

Excited States

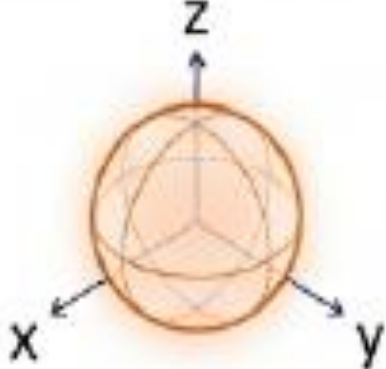
- **Consequence:** In bulk materials, energy bands are continuous. But in nanoscale structures, these bands split into discrete energy levels i.e. quantized.

Ground State

- **The Blue Shift:** When the size of the “box” decreases, the energy gap increases. That means smaller nanoparticles emit higher-energy (blue) light. Larger nanoparticles emit lower-energy (red) light.

- ***So, confinement changes energy structure — and energy structure determines material properties.***

The Dimensionality Matrix

Material Type	Quantum Well (2D)	Quantum Wire (1D)	Quantum Dot (0D)
	 Thin Films	 Carbon Nanotubes	 CdSe Nanocrystals
Confinement	1 Dimension (z)	2 Dimensions (y, z)	3 Dimensions (x, y, z)
Freedom	2 Dimensions (x, y)	1 Dimension (x)	0 Dimensions (Fixed)
Example	Thin Films	Carbon Nanotubes	CdSe Nanocrystals

Materials: How Size Rewrites Identity

- At small sizes, surface atoms dominate. In nanoparticles, a large fraction of atoms lie on the surface. These atoms have unsatisfied bonds, making the material more reactive and chemically active. Also, quantum confinement changes band structure. This modifies optical absorption and emission. So, size doesn't just reduce dimension — it changes identity.

GOLD



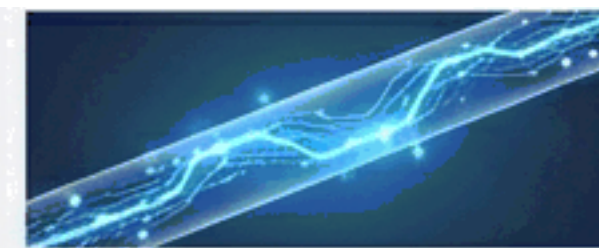
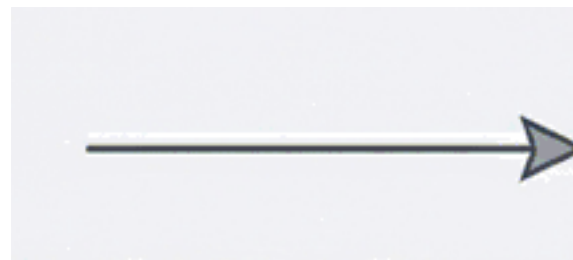
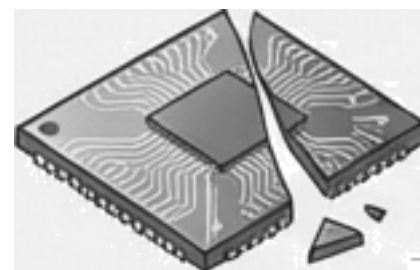
1. Bulk gold is yellow. But gold nanoparticles appear red or purple.

COPPER



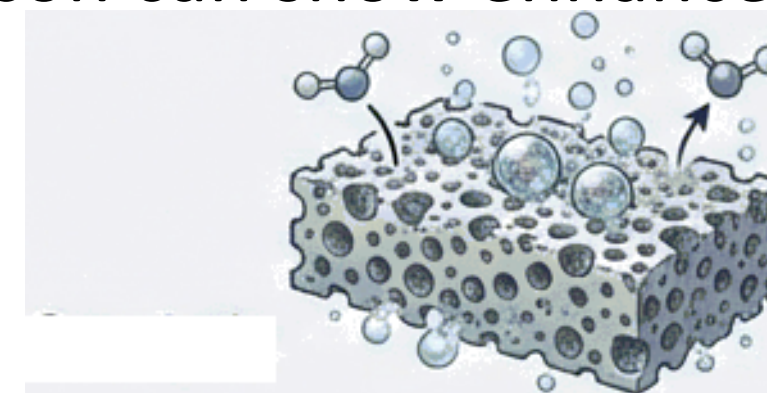
2. Bulk copper is opaque. Nano copper can become semi-transparent.

SILICON



3. Bulk silicon is an indirect semiconductor. Nano silicon can show enhanced luminescence.

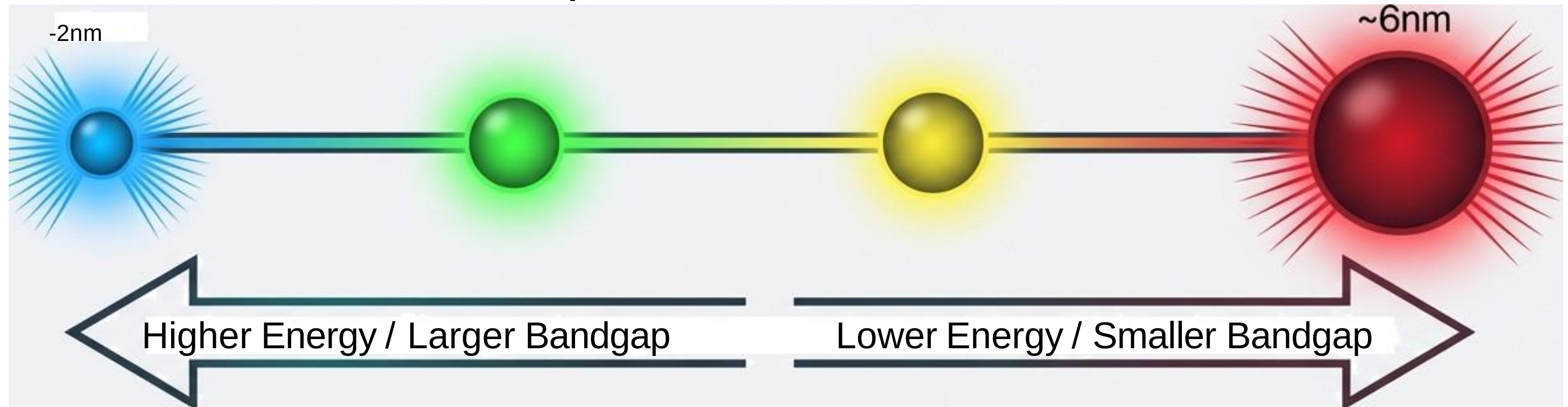
PLATINUM



4. Platinum nanoparticles become highly catalytic.

Zero Dimensions: The Tunable Quantum Dot

Spectral Size Chart



Quantum dots are 0D nanostructures where electrons are confined in all three directions. Because of complete confinement, their energy levels are fully discrete.

When light excites a quantum dot, it creates an electron-hole pair (exciton). When this pair recombines, it emits light.

The remarkable thing is:

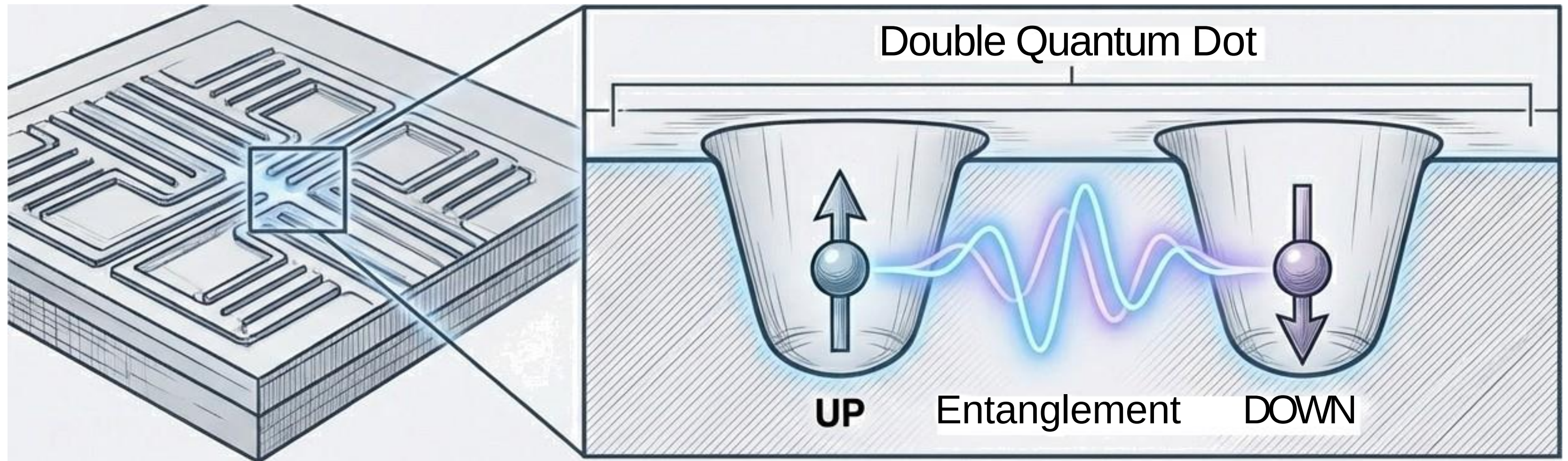
The color of emitted light depends only on size.

- **Small dots → large bandgap → blue emission**
- **Large dots → small bandgap → red emission**

This tunability makes quantum dots useful in:

A) LED displays (QLED TVs) (B) Bio-imaging (C) Lasers

The Computational Leap: Dots as Qubits



Quantum dots can trap single electrons. The electron's spin (up or down) can represent quantum information — similar to 0 and 1 but with quantum superposition.

Two nearby quantum dots can interact and become **entangled**. Entanglement allows correlated quantum states that classical systems cannot achieve.

Advantages of quantum dots:

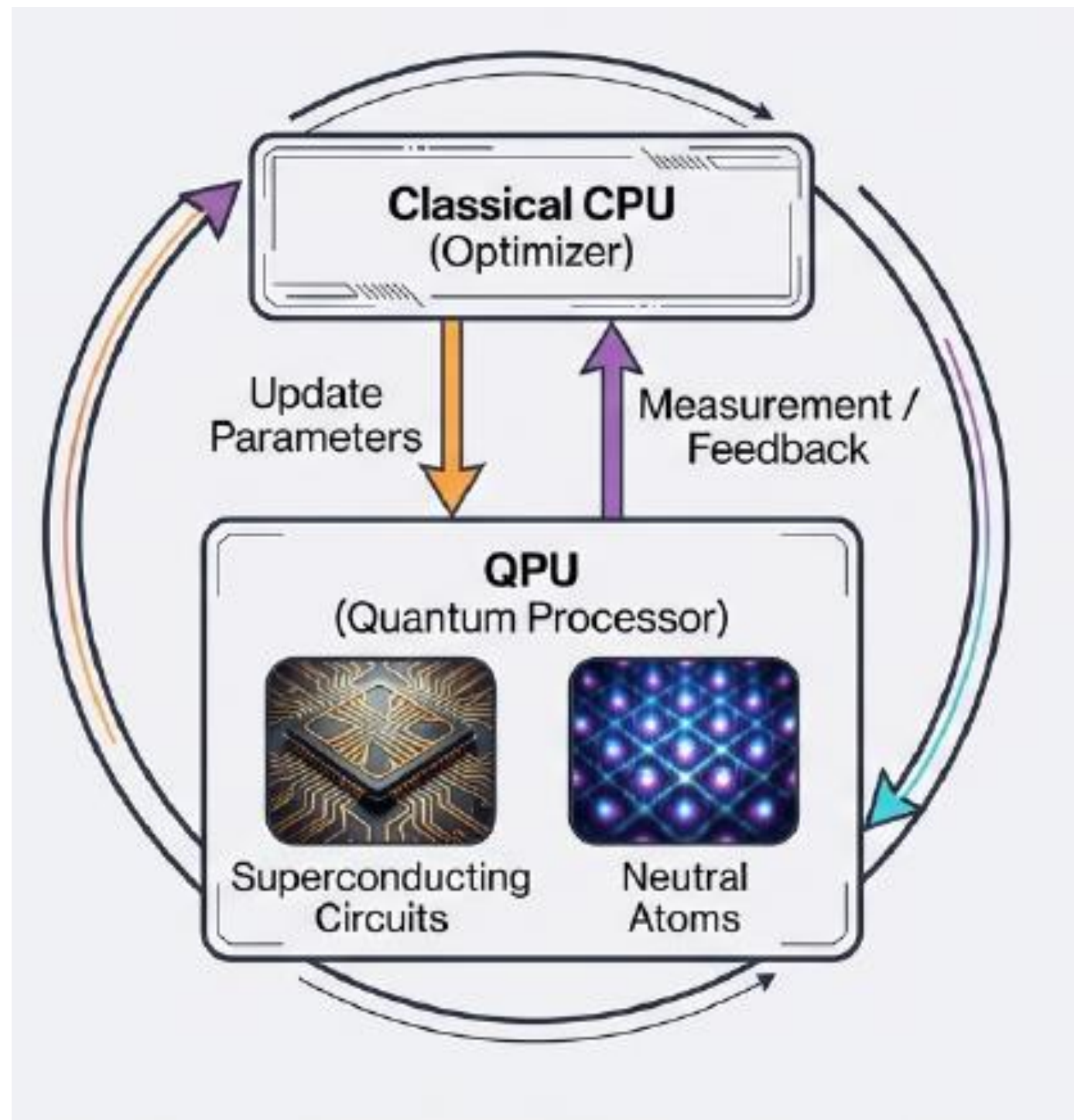
Solid-state platform (compatible with semiconductor industry)

Tunable energy levels

They are candidates for scalable quantum computers because we can fabricate many dots on a chip.

This connects nanomaterials directly to quantum computing hardware

Bridging the Gap: Middleware and Hybrid Architectures



Future quantum systems will be hybrid. Different qubit technologies will work together:

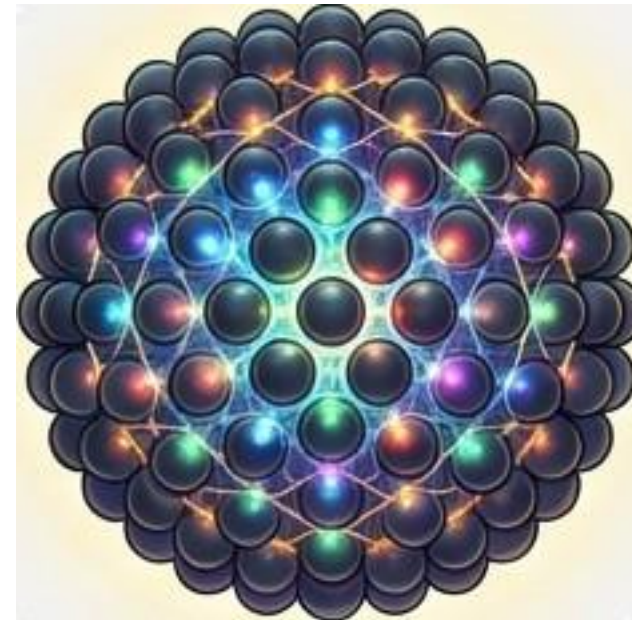
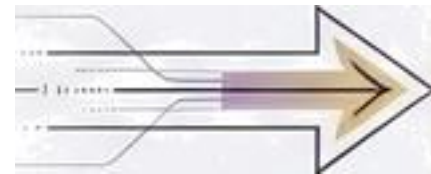
- Superconducting qubits (fast operations)
- Neutral atoms (flexible connectivity)
- Quantum dots (scalable semiconductor integration)

Classical computers (CPUs) control quantum processing units (QPUs). The QPU works like a special accelerator.

Hybrid systems reduce errors and improve performance.

This slide helps students see that quantum computing is not purely theoretical — it involves engineering integration of multiple nanoscale devices

The Future is confined



The central idea is:

- ❑ When we confine electrons, we gain control.
- ❑ Quantum confinement allows us to engineer:
Light emission, Conductivity, Spin states & Quantum logic
- ❑ From changing the color of gold to building quantum computers, all rely on controlling motion at nanoscales.

The philosophical takeaway:

“By restricting freedom of electrons, we gain freedom to design the future.”

This makes nanoscience one of the most powerful tools in modern engineering.